# Daniel Zhang

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## Education

**University of Waterloo** *Waterloo, ON*
PhD in Statistics; Supervisor: Prof. Ali Ghodsi; graduate average: 91% *Jan. 2025 – expected 2028*

- Research areas: Bayesian deep learning, variational inference, machine learning theory, uncertainty quantification, and efficient language models.
- Completed the first doctoral comprehensive examination through projects on Bayesian backpropagation-free neural networks and uncertainty-aware aggregation of stochastic LLM generations.

**University of Waterloo** *Waterloo, ON*
Master of Mathematics in Statistics; Supervisor: Prof. Ali Ghodsi; graduate average: 91% *Sept. 2023 – Aug. 2024*

- Research project: *GAM SymbolicGPT: A Generalized Additive Model Approach to Symbolic Regression*.

**University of Waterloo** *Waterloo, ON*
Honours Bachelor of Mathematics in Statistics; Minor in Computing; Co-operative Program *Sept. 2018 – Apr. 2023*

## Research Experience and Contributions

**University of Waterloo** *Waterloo, ON*
Doctoral Researcher, Data Analytics Laboratory; Supervisor: Prof. Ali Ghodsi *2025 – present*

- Research in Bayesian deep learning, variational inference, uncertainty quantification, and efficient language models.

**University of Waterloo** *Waterloo, ON*
Doctoral Comprehensive Research Projects; Supervisor: Prof. Ali Ghodsi *2025*

- **Bayesian NoProp:** introduced Bayesian linear and convolutional layers into a backpropagation-free learning framework, enabling decomposition of epistemic and aleatoric uncertainty through variational inference and Monte Carlo prediction.
- Studied KL annealing, label embeddings, calibration, and distribution shift on CIFAR-10/100; found that excessive KL regularization can induce posterior collapse and near-random performance in wider or harder settings.
- **Trust-Weighted LLM Aggregation:** formulated stochastic decoding as sampling over implicit reasoning paths and used accuracy- and embedding-variance-based trust scores to aggregate generations and assess entropy, calibration, and semantic disagreement.

**University of Waterloo** *Waterloo, ON*
Graduate Researcher, Data Analytics Laboratory; Supervisor: Prof. Ali Ghodsi *Sept. 2023 – Aug. 2024*

- Proposed *GAM SymbolicGPT*, extending neural symbolic regression with additive structure and a backfitting-inspired procedure to reduce the search burden in higher-dimensional problems.
- Designed experiments on synthetic and applied regression tasks to characterize where additive decomposition improves interpretability and where interactions limit the approach.

**Vision and Image Processing Laboratory, University of Waterloo** *Waterloo, ON*
Undergraduate Researcher; Supervisors: Prof. Alexander Wong, Dr. Jason Deglint, and Vikram Voleti *Sept. – Dec. 2022*

- Led *Plankton-FL*, a privacy-preserving federated learning study for phytoplankton classification across heterogeneous data sources; served as first author and presented the work at CVIS 2022.
- Investigated optimization, client participation, and model-alignment strategies, including Git Re-Basin, for learning across disjoint image domains.

**National Research Council Canada** *Toronto, ON*
Research Assistant; Supervisors: Dr. Mohammad Sajjad Ghaemi and Dr. James H. K. Ooi *Jan. – Apr. 2021*

- Developed a two-stage transfer-learning framework that pre-trained on influenza records and fine-tuned on COVID-19 clinical data for patient-severity prediction.
- Improved AUROC by 25% relative to an MLP trained from scratch and addressed class imbalance through clinical feature engineering and SMOTE.

## Publications

**Peer-reviewed publication**

**Zhang, D.**, Voleti, V., Wong, A., and Deglint, J. (2022). "Plankton-FL: Exploration of Federated Learning for Privacy-Preserving Training of Deep Neural Networks for Phytoplankton Classification." *Proceedings of the 8th Annual Conference on Computational Vision and Intelligent Systems*. **Best AI Paper Award**.

## Awards, Fellowships, and Distinctions

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| **2026** | Presentation Award, 6th Waterloo Student Conference in Statistics, Actuarial Science, and Finance, University of Waterloo. |
| **2026** | Statistics and Actuarial Science Chair's Graduate Award, University of Waterloo |
| **2024–2026** | Graduate Research Studentship, University of Waterloo |
| **2022** | Best AI Paper Award, 8th Annual Conference on Computational Vision and Intelligent Systems (CVIS 2022), Waterloo AI Institute |
| **2018** | President's Scholarship, University of Waterloo |

## Presentations and Selected Research Training

**Zhang, D.** (2026). "Tune Small, Train Big: BNN-$\mu$P for Bayesian Neural Network Language Models." Oral presentation, Waterloo Student Conference in Statistics, Actuarial Science and Finance, University of Waterloo, Waterloo, ON, October 23, 2026.

**Zhang, D.**, Voleti, V., Wong, A., and Deglint, J. (2022). "Plankton-FL: Exploration of Federated Learning for Privacy-Preserving Training of Deep Neural Networks for Phytoplankton Classification." Oral presentation, Computational Vision and Intelligent Systems Conference, Waterloo, ON.

Selected participant, *Princeton Machine Learning Theory Summer School*, Princeton University, Aug. 2026.

## Teaching and Academic Leadership

**University of Waterloo** *Waterloo, ON*
Head Teaching Assistant *2024, 2026*

- STAT 940 (Deep Learning) and STAT 442/842 (Data Visualization); supported course delivery, assessment coordination, and consistent feedback across undergraduate and graduate cohorts.

**University of Waterloo** *Waterloo, ON*
Teaching Assistant *2023 – 2026*

- Supported ten course offerings spanning statistical learning, advanced regression, case studies in data science, computational statistics, forecasting, data visualization, and introductory statistics.
- Developed grading guidance and detailed feedback for technical reports, case studies, presentations, and statistical analyses.

## Consulting and Applied Data Science Experience

**Statistical Consulting and Survey Research Unit** (**SCSRU**) *Waterloo, ON*
Statistical Consultant, University of Waterloo *Sept. 2026 - present*

- Provide statistical consulting on a variety of projects during office hours.
- Collaborate with a team on an interdisciplinary project.

**Intact Financial Corporation** *Toronto, ON*
Data Scientist I, Usage-Based Insurance *June 2023 – Dec. 2024*

- Analysed more than 500,000 telematics trips and over 100 candidate features; reduced the active feature set by 25% while improving the driving-risk model's Gini coefficient by 2%.
- Collaborated with data scientists, developers, and product teams on production driving-event logic and validation for mobile features used by more than 100,000 users.

**Intact Financial Corporation** *Toronto, ON*
Data Science Intern, Usage-Based Insurance *Sept. – Dec. 2022; Sept. – Dec. 2021*

- Built PySpark and Pandas pipelines for commercial-vehicle and insurance-pricing applications, integrating high-volume telematics data with claims and actuarial modelling workflows.
- Performed statistical feature screening, refactored production pipelines, and created modular preprocessing tools for custom actuarial features.

**TD Bank** *Toronto, ON*
Data Science and Advanced Analytics Intern *May – Aug. 2022*

- Automated legacy analytics workflows and developed SQL/PySpark data pipelines for TD Wealth; contributed to migration of ETL processes to Azure-based infrastructure.

## Technical Skills

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| **Research** | Bayesian neural networks, variational inference, uncertainty quantification, $\mu$P and scaling laws, Transformers, calibration, distribution shift |
| **Programming** | Python, PyTorch, NumPy, Pandas, scikit-learn, TensorFlow/Keras, R, SQL, PySpark, SAS |
| **Computing** | SLURM, multi-GPU and distributed training, Linux, Git, high-performance computing, LaTeX |